SHIELD: Failure-Aware Shielding for Vision-Language-Action Models

Daniel Contreras-Esquivel Tianhui (Alina) Huang Jacob Lee
Department of Computer Science, Stanford University

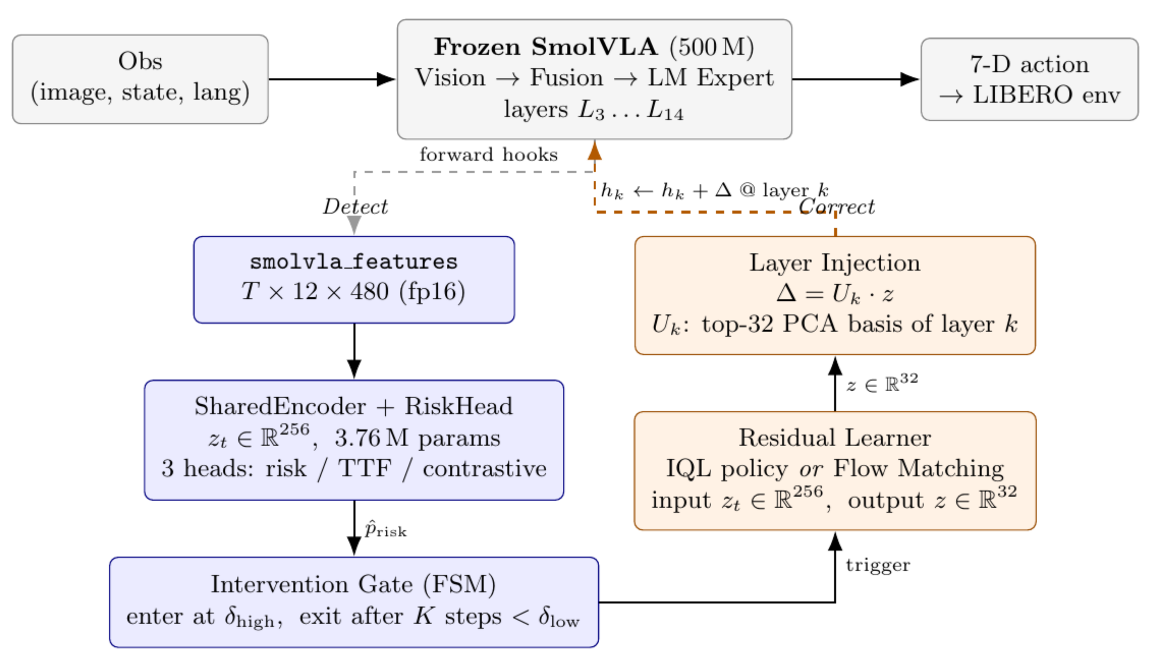
Problem, Prior Work & Architecture

Problem. VLA policies fail *silently* under distribution shift — sensor noise, viewpoint drift, and action jitter corrupt hidden states with no detect-or-correct loop.

SHIELD is the first *triggered*, *learned*, activation-space shield for VLAs: fires only when calibrated risk exceeds a threshold; injects a flow-model correction; falls back safely when inactive. No base weights modified.

Research question. Can we detect impending manipulation failures from VLA activations and inject a lightweight residual to recover success, without retraining the base model?

Prior work positioning. Activation steering (RepE, ITI, ROME) shows residuals can steer representations, but not continuous-action VLAs. VLA robustness work (VLA-OS, Cerebellar, ReSteer) improves execution, but is mostly action-space or not triggered. COAST uses conceptor steering; SHIELD adds calibrated risk gating and a learned flow residual.



Data & Perturbations

5,117 labeled rollouts of SmolVLA on LIBERO across 11 perturbation types. Label: `fail_in_H` ($H = 30$) + per-episode failure-type tag.

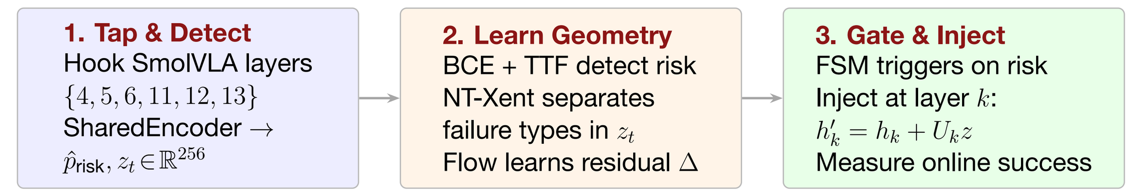
Split	Contents
Eval only	Joint-angle noise (hard negatives)
Train + eval	Action noise, viewpoint shift, Gaussian blur + windowed variants (×2 each)
Trigger retrain	999 windowed + 250 nominal successes (timeout failures excluded)
Held out	4,118 continuous perturbations

Labels. Binary `fail_in_H` ($H = 30$) + coarse failure-type tag (collision, drop, grasp misalignment, grasp precision) — the NT-Xent supervision signal for z_t geometry. Windowed perturbations give clean risk-on/off transitions per episode; nominal timeout failures excluded from retraining.

Why this split matters. Risk detection is trained on temporally local windows, while final evaluation is online and closed-loop: the policy must act under fresh perturbations, not just classify stored trajectories.

Research Paradigm

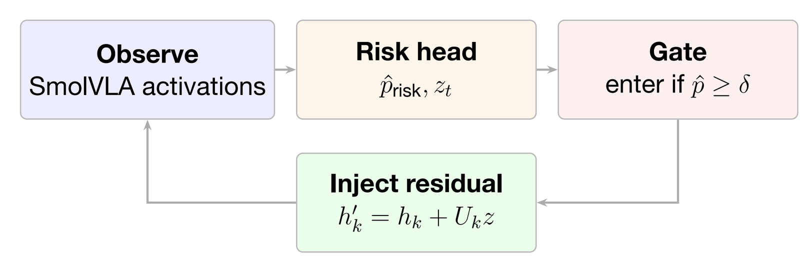
Hypothesis. Failure is encoded in VLA activations *before* it manifests in actions. A calibrated trigger plus a learned flow residual can redirect the policy back to success without modifying the base model.



Main result. On a hard action-noise task, SHIELD improves online success from **30% to 60%** without modifying base SmolVLA weights.

Success criterion. Shielded SR $>$ unshielded SR on perturbed episodes; clean SR must not degrade. Current demos show selective recovery: Task 3 $+30$ points, Task 4 $+10$ points, easy tasks flat/slightly worse under false positives.

Online control loop. SHIELD is dormant until calibrated risk crosses a threshold; then a finite-state machine injects a learned residual for a short recovery window and exits after risk stays low.



Without shield: risk can rise and persist until the rollout times out.

With shield: the trigger creates a bounded correction window; success depends on trigger precision.

Risk Head Architecture

SharedEncoder (3.76M params): temporal CNN over 6 hooked layers + cross-layer attention → two outputs:

- $\hat{p}_{\text{risk}} \in [0, 1]$ — calibrated failure prob
- $z_t \in \mathbb{R}^{256}$ — failure-type latent for flow conditioning

$$\mathcal{L} = \lambda_{\text{BCE}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{TTF}}\mathcal{L}_{\text{TTF}} + \lambda_{\text{NT-Xent}}\mathcal{L}_{\text{NT-Xent}}$$

FSM & injection. Enter: $\hat{p} \geq 0.7$; exit: $K = 5$ steps below 0.4. Flow model: $\Delta = U_k z$, $h'_k = h_k + \Delta$ via top-48 PCA basis. No VLA weights modified.

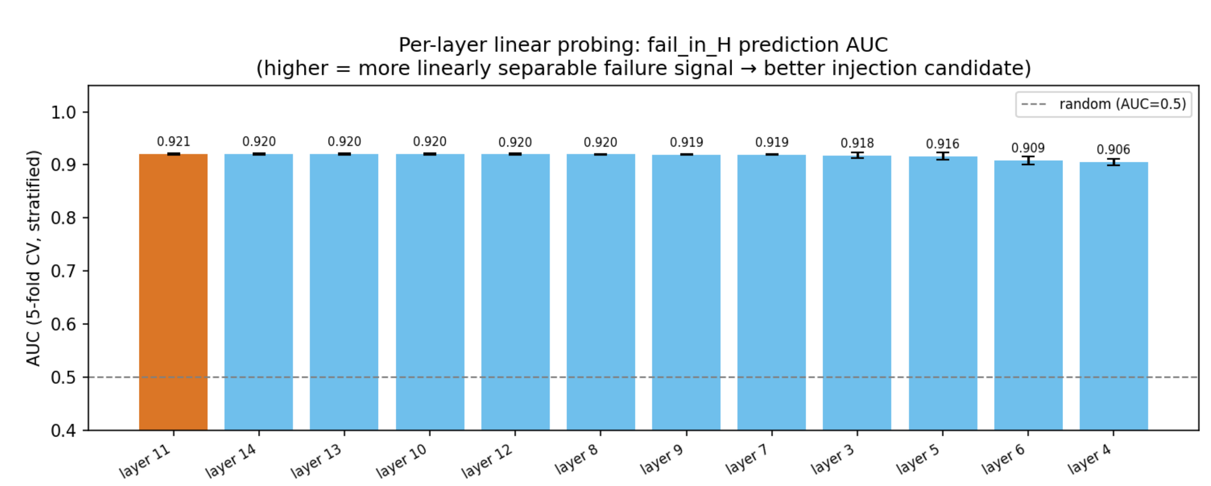


Figure 2. Failure risk is linearly readable across action-expert layers (AUC ≈ 0.91–0.92), so SHIELD can trigger before the action visibly fails.

Risk Head Ablation

Run	wAUC	Silh.	ECE	BSkill
A BCE only	0.928	−0.046	0.088	−0.067
B BCE+TTF	0.927	−0.040	0.093	−0.12
C BCE+TTF+NT-Xent	0.917	+0.168	0.077	+0.018
D NT-Xent only	0.440	+0.125	—	—

Table 1. Loss ablation (Run C selected, highlighted).

NT-Xent costs only -0.010 wAUC vs. B while improving calibration and moving the latent from mixed failure states to a typed geometry (silhouette $-0.040 \to +0.168$). We choose Run C because the downstream flow model needs *where to steer*, not just whether risk is high.

Phase 2 → Phase 3 Evidence

Paired rollout. Same task, seed, and perturbation: unshielded times out at 600 steps; B6 layer-11 injection succeeds in 80 steps.

Why this matters. The baseline is the same SmolVLA with no injection; the gain comes from the gated activation residual, not retraining or changing the policy weights.

Shield Evaluation

Current online demos use the retrained Run C risk head ($T = 1.34$) and B6 layer-11 conceptor-flow injection. The key pattern is task-dependent: the shield rescues harder action-noise rollouts, but over-triggers when the base policy is already strong.

Regime	Unsh.	SHIELD	ΔSR	Read
Task 3 action noise	0.30	0.60	+0.30	strongest rescue
Task 4 action noise	0.80	0.90	+0.10	smaller gain
Easy tasks 0–2	0.81	0.79	−0.02	over-trigger risk

Table 2. Demo-scale Phase 3 results, 10 episodes/task. Strongest evidence: task-matched rescue on Task 3.

Key diagnostic. B6 triggers on nearly every action-noise episode. This creates recoveries when the base policy is failing, but high FPR (Task 3 ≈ 0.91, Task 4 ≈ 0.66) can also cause timeouts from unnecessary intervention.

Interpretation. The residual is useful when the rollout is actually drifting toward failure; the remaining bottleneck is deciding *when not to intervene*. That is why thresholding and cooldown are the next controls, not a larger flow model.

Insights & Takeaways

1. Detection signal. Failure risk is readable before visible failure: layer probes reach AUC ≈ 0.92 across all action-expert layers.

2. Geometry matters. NT-Xent trades only -0.010 wAUC for silhouette $-0.040 \to +0.168$, giving the flow a typed failure latent to condition on.

3. Online recovery appears task-matched. B6 layer-11 conceptor flow improves Task 3 $0.30 \to 0.60$ SR and Task 4 $0.80 \to 0.90$ SR.

4. Main limitation. High-FPR always-trigger behavior can disturb easy rollouts; thresholds, cooldown, and routing are the next levers.

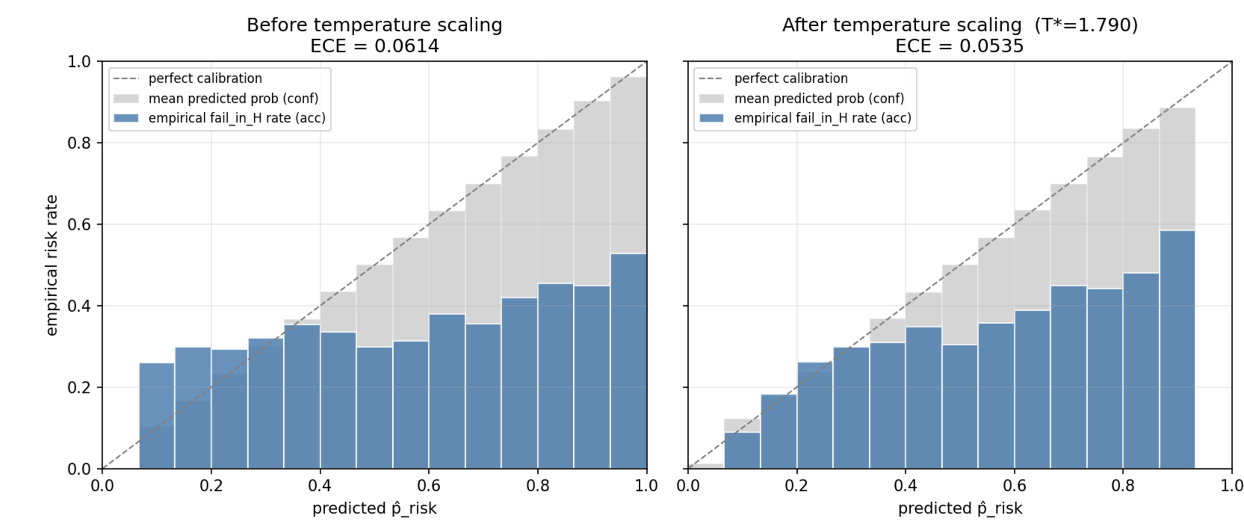


Figure 3. Run C (retrained) reliability diagram after temperature calibration ($T = 1.34$). ECE 0.040; Brier 0.053. Well-calibrated $\hat{p}_{\text{risk}}$ makes thresholded interventions interpretable rather than arbitrary.

Rubric-level takeaway. The project contribution is not only a detector; it is a detect–gate–correct system with evidence at three levels: offline risk geometry, flow residual learning, and online simulation behavior.

Limitations & Future Work

What remains unresolved. SHIELD works best when risk is real; the next version must reduce false interventions on easy rollouts while preserving the Task 3 rescue behavior.

Why this matters. The current system proves that activation residuals can recover some perturbed rollouts, but the trigger is still conservative: if risk stays high after the base policy has already recovered, the shield can over-correct and turn an otherwise solvable episode into a timeout. The remaining challenge is therefore *selective intervention*, not simply a larger flow model.

1. **Trigger precision.** Replace one fixed threshold with adaptive δ_{high}, cooldown, and risk-slope checks so SHIELD fires later on nominal rollouts.
2. **Failure-aware routing.** Select injection scale and layer by failure type: action noise, blur, and viewpoint shifts may need different residuals.
3. **Stronger evidence.** Evaluate full LIBERO suites with more seeds, then compare PCA, U_{null}, and U_{potent} to test whether recovery follows action-sensitive directions.

Phase 3 plan. Treat the online shield as a control policy: sweep threshold/cooldown to improve trigger selectivity, route residuals by failure type and layer, then validate on full LIBERO suites with clean-policy retention.

danielce | alinah | jacoblee @stanford.edu